

The inverse self-commutator cost at inertia $(m, 3)$: tiling certificates, 3×3 -block shifts and a padding phase

Lluis Eriksson (Independent researcher)

13 September 2026

Abstract

For a traceless Hermitian $F \in M_d(\mathbb{C})$ let $\kappa_d(F) = \frac{1}{2} \min\{\|C\|_{HS}^2 : CC^* - C^*C = 2F\}$. On the inertia- $(m, 3)$ stratum (m positive eigenvalues, three negative ones, z zeros) we determine κ_d as the maximum of an explicit finite set $S(m, z)$ of linear forms for $m = 3, \dots, 9$ and $z \leq 2$, by exact Horn “tiling” certificates below and 3×3 -block weighted shifts above, the latter verified at every vertex of every chamber together with a convexity argument. The forms are costs of layerings; for the ordered family A the lower bound is proved for all m and z by Lidskii–Wielandt tail inequalities. Unlike inertia $(m, 2)$, the value is not padding-invariant: for $m \equiv 1, 2 \pmod{3}$ exactly one extra exposed form exists at $z = 0$, so $\kappa_{m+3} > \kappa_{m+4}$ on an open set and rank $m + 1$ is forced after padding (exact example: $\kappa_7 = 74/61 > \kappa_8 = 73/61$). Six combinatorial families are conjectured to describe $S(m, z \geq 1)$ for all m , with quadratic growth; they are verified out of sample for $m = 10, 11$.

1. Introduction

The quantity $\kappa_d(F)$ measures the cheapest realisation, in Hilbert–Schmidt norm, of a traceless Hermitian F as a self-commutator $\frac{1}{2}(CC^* - C^*C)$. Its theory has been developed in a line of records of the AI Research Record archive, cited by descriptive keys (identifiers in the reference list); everything used from them is restated here. [FourLevel] reduced κ_d to a finite linear program over Horn inequalities (the *Horn program*, restated as (2.1)) and gave the four-level formula; [FiveLevel] the five-level formula; [OneSpike] the one-spike formula $\kappa_d = \sum_j j a_j$ when one sign has multiplicity one; [RankOnset] the rank-onset results and, in its Remark 3.3, the first example where zero padding lowers the value, $\lambda = (25, 18, 18, -10, -17, -17, -17)/7$ with $\kappa_7 = 74/7$ and $\kappa_8 = 73/7$. The immediate antecedent is the inertia- $(m, 2)$ paper [Mtwo], which proved a dimension-free closed formula $\kappa_d = \max(F_0, \dots, F_{m-1}, G_1, G_3, \dots)$ for every m and every number of zeros (Theorem A there), described its chambers (Theorem B) and optimizers (Theorem C’), and asked in its Open question 1 what happens at inertia $(m, 3)$: whether block shifts with 3×3 blocks attain the value and whether Horn triples adding two boxes to a rectangle certify it.

Contribution. We answer these questions on the inertia- $(m, 3)$ stratum for $m \leq 9$ and $z \leq 2$ zeros ($d \leq 14$) and describe what changes. (i) The value is again the maximum of finitely many linear forms, each the cost of a *layering* of the spectrum into layers of size at most three (Theorem 1), and the number of exposed forms grows quadratically in m (10, 18, 26, 34, 45, 59, 69 for $m = 3, \dots, 9$) instead of linearly. (ii) Every form is certified below by a sum of Horn inequalities with multipliers one, each s_t used at most once — a “tiling” — built from Lidskii–Wielandt inequalities and Pieri triples with $\lambda(j)$ a single

row or column of at most three boxes; for the ordered family A the certificate is explicit and dimension-free (Theorem 2, proved for all m, z). (iii) The value is attained by 3×3 -block weighted shifts, verified as a linear program at every vertex of every chamber and extended to the chambers by convexity. (iv) Padding is *not* invariant: when $m \equiv 1, 2 \pmod{3}$ there is exactly one extra exposed form at $z = 0$ which is invalid after padding, so $\kappa_{m+3} > \kappa_{m+4}$ on an open set, and after padding every optimizer there has rank $m + 1$ (Theorem 3; the padding example of [RankOnset] is the $m = 4$ instance). (v) The exposed forms for $z \geq 1$ are conjecturally the costs of six explicit families A–F of layerings, verified for $m \leq 9$ against the computed sets and out of sample for $m = 10, 11$ (Conjecture 4).

Status, stated honestly. Theorem 1 is proved with computer assistance for $m = 3, \dots, 9, z = 0, 1, 2$; Section 3 states, for each (m, z) , which half of the proof is exact arithmetic and which is floating point. Theorem 2 is proved for all m and z . Theorem 3 is proved for $m \leq 9$ (with exact rational strict inequalities at explicit points) and observed numerically for $m = 10, 11, 12, 13$. Conjecture 4 is a conjecture. The manuscript was subjected to an independent adversarial review that re-implemented every computational ingredient and upgraded the upper bound to exact arithmetic for $m \leq 8$ (extended to $m = 9$ with its code for this manuscript); its corrections are incorporated (Section 8). Two literature searches (Section 10) found no prior closed formula on an inertia stratum and no prior block-shift or tiling-certificate method; the nearest works, Johnson–Ozawa–Schechtman [JOS13] and Angel–Schechtman [AS15], bound $\|B\| \|C\|_2$ over factorisations $A = [B, C]$, a different objective. A null search does not certify priority.

2. Setting

Throughout $\|X\|^2 = \text{Tr } X^*X$ and $F \in M_d(\mathbb{C})$ is Hermitian, traceless, with ordered spectrum $\lambda_1 \geq \dots \geq \lambda_d$. On the inertia- $(m, 3)$ stratum

$$\lambda = (a_1 \geq \dots \geq a_m > 0, 0^z, -b_3, -b_2, -b_1), \quad b_1 \geq b_2 \geq b_3 > 0, \quad \sum_j a_j = b_1 + b_2 + b_3 =: P, \quad d = m + z + 3.$$

The spectrum index of a_j is j ; those of $-b_1, -b_2, -b_3$ are $d, d-1, d-2$. Since every statement is homogeneous we normalise $P = 1$ when convenient. By $F \mapsto -F$ (which replaces C by C^*) everything transfers to inertia $(3, n)$.

Given C with $CC^* - C^*C = 2F$, put $R = CC^*/2, S = C^*C/2$: they are PSD with the same spectrum $s_1 \geq \dots \geq s_d \geq 0$ (the *common spectrum*), $R - S = F, \frac{1}{2}\|C\|^2 = \sum_t s_t$, and conversely any such pair comes from $C = U\sqrt{2\Sigma}V^*$. Hence ([FourLevel] Theorem 2.2, [RankOnset] Proposition 2.2, [Mtwo] (2.1))

$$\kappa_d(F) = \min \left\{ \sum_{t < d} s_t : s_1 \geq \dots \geq s_{d-1} \geq s_d = 0, (s, -s^{\text{rev}}, \lambda) \text{ Horn-feasible} \right\}, \quad (2.1)$$

where Horn-feasible means that $\alpha = (s_1, \dots, s_{d-1}, 0), \beta = (0, -s_{d-1}, \dots, -s_1)$ and $\gamma = \lambda$ are the spectra of Hermitian $R, -S$ and $F = R + (-S)$, i.e. satisfy the trace identity and every Horn inequality $\sum_{k \in K} \gamma_k \leq \sum_{i \in I} \alpha_i + \sum_{j \in J} \beta_j, (I, J, K) \in T_r^d$ [Ho62, Kl98, KT99, Fu00]. In s -notation a triple reads

$$\sum_{i \in I, i < d} s_i - \sum_{j \in J, j > 1} s_{d+1-j} \geq \sum_{k \in K} \lambda_k. \quad (2.2)$$

With $\lambda(I) = (i_r - r, \dots, i_1 - 1)$ the partition attached to $I = (i_1 < \dots < i_r)$, (2.2) holds for all Hermitian triples whenever the Littlewood–Richardson coefficient $c_{\lambda(I)\lambda(J)}^{\lambda(K)}$ is nonzero (Klyachko [Kl98]); this

necessity direction is all the lower bounds use. Sufficiency (Klyachko, Knutson–Tao) enters twice: in dimensions $n \leq 3$ inside the block-shift construction (Section 5), and in identifying the set of optimal common spectra with the optimal face of (2.1) (Section 6). Two facts about $\lambda \mapsto \kappa_d(\lambda)$ are used repeatedly: it is the value of a linear program in which λ enters only the right-hand sides, hence convex and piecewise linear in λ (so $\kappa_d - g$ is convex for every linear form g , and κ_d is continuous on the closed stratum); and $\kappa_{d+1}(F \oplus 0) \leq \kappa_d(F)$ (pad C by a zero row and column).

Forms and canonical coordinates. A *form* is a linear function $g(\lambda) = \sum_j \alpha_j a_j + \beta_1 b_1 + \beta_2 b_2$ with integer coefficients, written $(\alpha_1, \dots, \alpha_m; \beta_1, \beta_2)$; the coefficient of b_3 is normalised to 0 using $\sum a = \sum b$. A form is *valid* if $g \leq \kappa_d$ on the stratum and *exposed* (relative to a set S of valid forms) if it is the unique maximiser of S on a nonempty open subset. Its *chamber* is $C_g = \{\lambda \text{ in the stratum, } P = 1 : g \geq h \forall h \in S\}$, a polytope.

Layerings and costs. A *layering* L is an ordered partition of the multiset $\{-b_1, -b_2, -b_3, a_1, \dots, a_m\}$, possibly together with some of the zero eigenvalues, into layers $\Lambda_0, \Lambda_1, \dots, \Lambda_T$ of size at most 3, with $\Lambda_0 \subseteq \{-b_i\}$ and Λ_T a nonempty set of a 's (and zeros). Write $t(x)$ for the index of the layer containing x , $\text{tail}_u = \sum_{x \in \Lambda_u \cup \dots \cup \Lambda_T} x$, and

$$\text{cost}(L) = \sum_{u=1}^T \text{tail}_u.$$

Lemma 2.1. $\text{cost}(L) = \sum_x t(x) x = \sum_j (t(a_j) - t(b_3)) a_j + \sum_i (t(b_3) - t(b_i)) b_i$, where $t(b_i) := t(-b_i)$.

Proof. $\sum_{u \geq 1} \text{tail}_u = \sum_x x \cdot \#\{u \geq 1 : u \leq t(x)\} = \sum_x t(x) x$; adding $t(b_3)(\sum_i b_i - \sum_j a_j) = 0$ gives the canonical coordinates. $\square$

Conversely a form g with $\alpha_j = t(a_j) - t(b_3)$, $\beta_i = t(b_3) - t(b_i)$ for some nonnegative layer indices determines its *level layering* $L(g)$: the layers are the level sets of t , ordered by t , with the common shift fixed by $\min t = 0$. Every exposed form found below has a level layering with layers of size at most three, Λ_0 consisting of b 's and Λ_T of a 's. The *b-schedule* of a layering is $u = (t(b_1), t(b_2), t(b_3))$ and the *capacity* of layer $u \geq 1$ is $c_u = \#\{i : t(b_i) < u\}$.

3. Statements

Let $S(m, z)$ be the set of exposed forms computed in Section 8 (files `closed_m{m}_z{z}.json`; the sets for $z = 1, 2$ coincide, and for $z = 3$ when $m \leq 6$). Their sizes are:

m	3	4	5	6	7	8	9
$ S(m, 0) $	10	19	27	34	46	60	69
$ S(m, z \geq 1) $	10	18	26	34	45	59	69

(For $m = 2$ both sets have 5 elements, matching the count $(m - 1) + \lceil m/2 \rceil + \lceil m \geq 3 \rceil = 5$ of Theorem B of [Mtwo] at inertia $(3, 2)$.)

Theorem 1 (PROVED-COMPUTER-ASSISTED; $m = 3, \dots, 9, z = 0, 1, 2, d \leq 14$). On the whole stratum $\kappa_d = \max_{g \in S(m, z)} g$. Every $g \in S(m, z)$ is the cost of its level layering $L(g)$, and $\kappa_d = g$ on the chamber C_g ; the chambers cover the stratum. The proof has two finite halves whose arithmetic status is:

(m, z)	d	lower bound $\kappa_d \geq g$	upper bound $\kappa_d \leq g$ on C_g
$m \leq 7, z \leq 2; (8, 0),$ $(8, 1); (9, 0)$	≤ 12	exact tiling certificates (Section 4)	exact: rational vertices + exact chain LP (reviewer's code; $m \leq 8$ in the review, $m = 9$ rerun for this manuscript)
$(8, 2); (9, 1), (9, 2)$	13; 13, 14	floating-point validity LP	exact (as above)

“Exact tiling certificate” means: a list of Horn triples, each with LR coefficient ≥ 1 verified by two independent implementations (and membership in Fulton’s T_r^d for $d \leq 9$), whose s -forms have every coefficient ≤ 1 and whose right sides sum, in integer arithmetic, to g . “Floating-point validity LP” means the joint linear program $\min_{\lambda}(\kappa_d - g) \geq -10^{-9}$ over the $P = 1$ stratum (HiGHS). The upper bound is, in all cases, the feasibility of one 3×3 -block chain per chamber at every chamber vertex plus convexity (Section 5); the author’s runs of it are floating point (qhull vertices, HiGHS chain LP) for all 787 chambers and 8 775 vertices, the reviewer’s exact (pycddlib/GMP) for $m \leq 8$, and the reviewer’s code rerun at $m = 9$ for this manuscript is exact as well, so the upper bound is exact for every (m, z) of the theorem.

Theorem 2 (PROVED, all $m \geq 3, z \geq 0$). Let L be a layering of family A: b -schedule $(0, u_2, u_3)$ with $u_2 \leq u_3$, and $a_1, a_2, \dots$ placed in order into the layers $u = 1, 2, \dots$, each layer $u < T$ receiving exactly c_u of them, the last layer T receiving the rest (at least one). Then $\kappa_d \geq \text{cost}(L)$ on the stratum, with an explicit Lidskii–Wielandt certificate (Section 4.2). In particular $\kappa_d \geq \max_A \text{cost}$ in every dimension.

Theorem 3 (padding and rank; PROVED-COMPUTER-ASSISTED for $m \leq 9$, exact where stated).

1. For $m = 3, 6, 9$: $S(m, 0) = S(m, 1) = S(m, 2)$ (and $= S(m, 3)$ for $m = 3, 6$), so $\kappa_{m+3}(\lambda) = \kappa_{m+4}(\lambda \oplus 0) = \kappa_{m+5}(\lambda \oplus 0^2)$ on the stratum.
2. For $m = 4, 5, 7, 8$: $S(m, 0) = S(m, 1) \cup \{g_0^{(m)}\}$ with exactly one extra form,

$$g_0^{(4)} = (1, 2, 3, 4; -2, -1), \quad g_0^{(5)} = (1, 2, 3, 3, 4; -2, -1),$$

$$g_0^{(7)} = (1, 2, 2, 3, 3, 4, 5; -2, -1), \quad g_0^{(8)} = (1, 2, 2, 3, 3, 4, 4, 5; -2, -1),$$

the costs of the layerings

$$\{b_3\}\{b_2, a_1\}\{b_1, a_2\}\{a_3\}\{a_4\}, \quad \{b_3\}\{b_2, a_1\}\{b_1, a_2\}\{a_3, a_4\}\{a_5\},$$

$$\{b_3\}\{b_2, a_1\}\{b_1, a_2, a_3\}\{a_4, a_5\}\{a_6\}\{a_7\}, \quad \{b_3\}\{b_2, a_1\}\{b_1, a_2, a_3\}\{a_4, a_5\}\{a_6, a_7\}\{a_8\}.$$

This form is invalid for every $z \geq 1$, and $\kappa_{m+3}(\lambda) > \kappa_{m+4}(\lambda \oplus 0)$ on a nonempty open subset of the stratum: at the rational points of Section 6 the exact margins $g_0 - \max S(m, 1)$ are $1/40, 1/60, 1/120, 1/150$. The value is padding-invariant for $z \geq 1$ within the verified range ($S(m, 1) = S(m, 2); = S(m, 3)$ for $m \leq 6$).

3. (Exact, GMP.) At $a = (17, 17, 17, 10)/61$, $b = (25, 18, 18)/61$ ($m = 4$), $\kappa_7 = 74/61$ with an optimizer of rank $4 = m$, while $\kappa_8 = 73/61$ and every optimizer of the padded problem has rank $\geq 5 = m + 1$: the minimum of s_5 over the optimal face of (2.1) is $1/61$, and the value of (2.1) with $s_5 = s_6 = s_7 = 0$ imposed is $74/61 = \kappa_7$. After $F \mapsto -F$ and scaling by $61/7$ this is the example of [RankOnset] Remark 3.3.

4. (Numerical.) The same pattern continues: at $z = 0$ exactly one extra exposed form appears for $m = 10, 11, 13$ (namely $(1, 2, 2, 3, 3, 3, 4, 4, 5, 6; -2, -1)$, $(1, 2, 2, 3, 3, 3, 4, 4, 5, 5, 6; -2, -1)$, $(1, 2, 2, 3, 3, 3, 4, 4, 4, 5, 5, 6, 7; -2, -1)$), each invalid at $z = 1$ by margins 0.033, 0.030, 0.026, and none for $m = 12$ in 600 samples; on the region where $g_0 > \max S(m, 1)$ every padded optimizer has rank $m + 1$ at 48/48, 40/40, 40/40 sampled points ($m = 4, 5, 7$), where the rank- m -capped padded value equals κ_{m+3} .

Conjecture 4 (general $m, z \geq 1$). $S(m, z \geq 1)$ is the set of costs of the layerings of the six families below, and $\kappa_d = \max$ over them for all $m \geq 3, z \geq 1$. Fix a b -schedule u with $\min u = 0$; layer $u \geq 1$ has c_u slots; concatenate the slots of layers 1, 2, 3, ... and fill them, left to right, with an arrangement π of $(a_1, \dots, a_m, *, *, *)$, where $*$ are virtual entries (value 0, deleted at the end, so that the last layers may be partial); a *boundary* is the position between the last slot of a layer and the first of the next. Moves: T_β transposes the two slots adjacent to boundary β ; F_β is the forward 3-cycle $(x, y \mid z) \mapsto (y, z \mid x)$ on the two slots before and the one after β ; B_β the backward 3-cycle $(x \mid y, z) \mapsto (z \mid x, y)$. Two rules apply to every family: the number of inversions of π (virtual entries counting as $+\infty$) equals the number of inversions of the b -order, $\#\{i < j : t(b_i) > t(b_j)\}$, and the last layer contains no b . The families are:

- **A**(u_2, u_3): $u = (0, u_2, u_3)$, $u_2 \leq u_3$, $\pi = \text{identity}$; exposed iff $2u_3 - u_2 \leq m - 1$ ($u_2 \geq 1$) or $2u_3 \leq m - 1$ ($u_2 = 0$), $u_2 \leq m - 2$, and $m \geq 4$ when $(u_2, u_3) = (0, 0)$. Count $A(m) = \lfloor \frac{m-1}{2} \rfloor + 1 - [m = 3] + (m - 2) + \lfloor \frac{(m-2)^2}{4} \rfloor$.
- **B**($u_2; \beta$): $u = (0, u_2, u_2 - 1)$, $u_2 \geq 1$ (b_3 one layer before b_2), one transposition T_β at a boundary ending a layer of index $\geq u_2$. Count $B(m) = \sum_{j=0}^{m-2} (\lfloor j/3 \rfloor + 1)$.
- **C**($k; \beta$): $u = (1, 0, k)$, $k \geq 1$ (b_2 alone in Λ_0 , b_1 in layer 1), one transposition at any boundary.
- **D**: $u = (1, 0, 0)$ ($\Lambda_0 = \{b_2, b_3\}$, $\Lambda_1 = \{b_1, a, a\}$): two transpositions at distinct boundaries, or one F_β .
- **E**: $u = (1, 1, 0)$ ($\Lambda_0 = \{b_3\}$, $\Lambda_1 = \{b_1, b_2, a\}$): two transpositions at distinct boundaries, or one B_β .
- **F**: $u = (2, 1, 0)$: exactly the two arrangements $(a_3, a_1, a_4, a_2, a_5, a_6, \dots)$ and $(a_2, a_3, a_4, a_1, a_5, a_6, \dots)$, i.e. $\{b_3\}\{b_2, a_3\}\{b_1, a_1, a_4\}\{a_2, a_5, a_6\} \dots$ and $\{b_3\}\{b_2, a_2\}\{b_1, a_3, a_4\}\{a_1, a_5, a_6\} \dots$

The generator (families2.py) reproduces $S(m, 1)$ exactly for $m = 3, \dots, 9$ with family counts A/B/C/D/E/F = 2, 5, 8, 11, 15, 19, 24 / 2, 3, 5, 7, 9, 12, 15 / 2, 4, 5, 8, 10, 14, 16 / 1, 2, 3, 3, 4, 6, 6 / 1, 2, 3, 3, 5, 6, 6 / 2, and predicts $|S(m)| = 87, 103, 120, 139, 164$ for $m = 10, \dots, 14$ (quadratic growth, against the linear count of inertia $(m, 2)$). Evidence beyond $m \leq 9$: at $m = 10$ and $m = 11$, $z = 1$, the predicted sets pass random tests, validity and exposedness LPs and a chamber-vertex completeness check (Section 8), which is a floating-point proof of $\kappa_{14} = \max$ of the 87 forms and $\kappa_{15} = \max$ of the 103 forms. What is not known: that the six families are exhaustive for general m , a general- z proof of the Pieri tiles of families B–F, and any general- m upper bound. Appendix A lists every form of $S(m, z)$ with its family and layering for $m \leq 6$.

4. Lower bounds

4.1 Tiling certificates

A *certificate* for a form g in dimension d is a list of Horn triples (I, J, K) (each with $c_{\lambda(I)\lambda(J)}^{\lambda(K)} \geq 1$) whose s -forms (2.2) sum to a linear combination of $s_1, \dots, s_{d-1}$ with every coefficient ≤ 1 and whose right sides sum to g . Since $s_t \geq 0$, $\sum_{t < d} s_t$ minus the summed left side is a nonnegative combination of the s_t , so every feasible s of (2.1) has $\sum_{t < d} s_t \geq g$; an optimizer with $s_d > 0$ is shifted as in [Mtwo] Corollary 4.6. Only Klyachko necessity is used, so a certificate is a proof of $\kappa_d \geq g$ for that d .

All certificates here are *tilings*: the multipliers are 1 and each s_t with $t \leq m$ occurs exactly once with coefficient +1; the only negative terms are $-s_{d-1}$, $-s_{d-2}$ (and occasionally $+s_{m+1}$, $+s_{m+2}$ on zero indices). Each triple is a *tile* of the form $+s[U] - s[N] \geq \sum_{k \in K} \lambda_k$ with $U \subseteq \{1, \dots, m+2\}$, $N \subseteq \{d-2, d-1\}$; the right side is a sum of consecutive a 's with at most three holes, minus some b 's. For every (m, z) with $d \leq 12$ the certificates were found as the duals of a *restricted* Horn LP over a catalogue of such tiles (`catalogue.py`, `certs.py`), solved at the strict Chebyshev centre of each chamber, then checked exactly: the multipliers rounded to 1 (they are within 10^{-6} of 1; what is verified exactly is the integer identity), the tiles' right sides sum in integer arithmetic (using $\sum a = \sum b$) to g , every s_t has total coefficient ≤ 1 , and every tile has LR coefficient ≥ 1 by two independent implementations (a tableau counter `lr.py` and the bialternant/Kostka formula `lr2.py`, cross-checked on 3000 random triples) and, for $d \leq 9$, lies in Fulton's recursively generated T_r^d . The reviewer re-verified 36 certificates (149 tiles) with a third LR implementation (Jacobi-Trudi + Pieri) and an own Fulton generator: 36/36.

The tile types at $z \geq 1$, by $\lambda(J)$, are: $\lambda(J) = \emptyset$ (Lidskii-Wielandt, including the Weyl tiles $s_j \geq a_j$ and $s_1 \geq b_1$); $\lambda(J) = (1)$ (the lower Weyl tile $s_2 \geq b_2$ and one-box Pieri tiles); $\lambda(J) = (1, 1)$ ($s_3 \geq b_3$ and two-box vertical strips); $\lambda(J) = (2)$ and (3) (horizontal strips). Counts per $(m, z = 1)$ for $m = 3, \dots, 8$: Lidskii-Wielandt 17, 40, 70, 110, 161, 232; row-type 8, 16, 24, 33, 50, 71; column-type 5, 8, 12, 12, 16, 22 (and 307/87/22 at $m = 9, z = 0$). At $z = 0$ the extra form $g_0^{(m)}$ needs, in addition, one or two tiles with $\lambda(J) = (3, 1)$ ($m = 4, 7$) or $(2, 1, 1)$ ($m = 5, 8$) — the only non-Pieri shapes in the whole computation. The highest positive s -index used is $m + 1$ ($z = 0$) or $m + 2$ ($z \geq 1$), never $d - 1$.

4.2 Family A: proof of Theorem 2

Let L be as in Theorem 2, with layers $\Lambda_0, \dots, \Lambda_T$. For $u = 1, \dots, T$ let $B_u \subseteq \{d-2, d-1, d\}$ be the spectrum indices of the b 's in layers $< u$ (so $|B_u| = c_u$), and t_u the index of the first a in layers $\geq u$; because the a 's are placed in order and every layer $u < T$ holds exactly c_u of them, the a 's of layer u are $a_{t_u}, \dots, a_{t_u+c_u-1}$ and $t_{u+1} = t_u + c_u$.

Tile u for $u < T$: the Lidskii-Wielandt triple $I = K = \{t_u, \dots, d\} \setminus B_u$, $J = \{1, \dots, r\}$, $r = |K| = d - t_u + 1 - c_u$. Here $\lambda(J) = 0$ and $\lambda(I) = \lambda(K)$, so $c = 1$; the sum condition $\sum I + \sum J = \sum K + r(r+1)/2$ is automatic. Its right side is $\sum_{k \in K} \lambda_k = \sum_{j \geq t_u} a_j - \sum_{i: t(b_i) \geq u} b_i = \text{tail}_u$ (the zeros contribute nothing). Its left side is

$$\sum_{i \in [t_u, d-1] \setminus B_u} s_i - \sum_{i=d+1-r}^{d-1} s_i = s_{t_u} + \dots + s_{t_u+c_u-1} - \sum_{\beta \in B_u, \beta < d} s_\beta,$$

because $d + 1 - r = t_u + c_u$ and the block $[t_u, t_u + c_u - 1]$ lies in $[1, m - 1]$, hence misses $B_u \subseteq \{d-2, d-1, d\}$.

Tile(s) for $u = T$: let $q = m - t_T + 1 \geq 1$ be the number of a 's in the last layer. If $t_T + c_T - 1 \leq m + z$ use the same Lidskii-Wielandt triple with $u = T$: right side $\sum_{j \geq t_T} a_j = \text{tail}_T$ (all b 's are in B_T), left side $s_{t_T} + \dots + s_{t_T+c_T-1} - \sum_{\beta \in B_T, \beta < d} s_\beta$, whose positive block may run into the zero indices $m + 1, \dots, m + z$ but not into B_T . Otherwise use the Weyl tiles $s_j \geq a_j$, $j = t_T, \dots, m$ ($I = K = \{j\}$, $J = \{1\}$), whose right sides also sum to tail_T .

Summation: the positive blocks of the tiles are the consecutive, pairwise disjoint index sets $[t_u, t_{u+1} - 1]$ ($u < T$) and $[t_T, \cdot]$; the negative terms involve only s_{d-1}, s_{d-2} . Hence every s_t , $t < d$, has total coefficient ≤ 1 , and $\sum_u \text{tail}_u = \text{cost}(L)$ by definition. By Section 4.1, $\kappa_d \geq \text{cost}(L)$. Nothing depends on d except through the index bookkeeping, which is valid for every $z \geq 0$. $\square$

The certificate is implemented LP-free in `familyA_cert.py` and verified exactly (sum condition, $LR = 1$ by `lr2`, disjointness, integer identity of the form) for all 945 (form, z) pairs with $m \leq 12$, $z \leq 4$. The reviewer re-derived the s -form and the identity independently. Its docstring mentions separate Weyl tiles for the b 's in Λ_0 ; the code emits none, because the $u = 1$ tile with $B_1 = \{d\}$ is $s_1 \geq b_1$ (and with $B_1 = \{d, d-1\}$ it is $s_1 + s_2 - s_{d-1} \geq b_1 + b_2$), exactly as described above. The forms F_k of [Mtwo] are the specialisation $b_3 = 0$ of family A. The argument uses only that every non-final layer is at full capacity, not $u_1 = 0$; but only the family-A layerings are exposed.

4.3 The Pieri tiles of families B–F

For families B–F the certificates are not dimension-free by inspection, but their tiles have only five shapes of $\lambda(J)$, and for each tile with $\lambda(J)$ a row (k) or a column (1^k) the Pieri rule [Fu97] gives $c_{\lambda(I)\lambda(J)}^{\lambda(K)} = 1$ if $\lambda(K)/\lambda(I)$ is a horizontal (vertical) strip of k boxes and 0 otherwise. So a general- z proof for a family reduces to two conditions per tile — the sum condition and the strip condition — as functions of the tail length; this was not carried out. Representative certificates at $m = 5$, $z = 1$ ($d = 9$), one per family, read as follows (right sides in a, b ; $s_1 \geq b_1$ is written as $s_1 \geq P - b_2 - b_3$ etc.):

family, form	tiles (left side $\geq$ right side)	$\lambda(I), \lambda(J), \lambda(K)$ of the non-Lidskii tile
B, $(-2, -1, 0, 1, 2; 3, -1)$	$s_5 \geq a_5;$ $s_4 + s_6 - s_8 \geq a_4 + a_5 - b_2;$ $s_3 \geq a_3 + a_4 + a_5 - b_3 - b_2;$ $s_2 \geq a_2 + \dots + a_5 - b_3 - b_2;$ $s_1 \geq P - b_3 - b_2$	$(3^4), (1), (4, 3, 3, 3)$
C, $(-2, -1, -1, 0, 1; 2, 3)$	$s_5 \geq a_5;$ $s_3 + s_6 - s_8 \geq a_4 + a_5 - b_3;$ $s_1 + s_4 - s_8 \geq a_2 + \dots + a_5 - b_3;$ $s_2 \geq P - b_3 - b_1$	$(3, 3, 3, 2), (1), (3^4);$ $(1^5), (1), (1^6)$
D, $(1, 1, 2, 3, 2; -1, 0)$	$s_4 \geq a_4;$ $s_1 + s_5 + s_6 - s_7 - s_8 \geq a_3 + a_4 + a_5;$ $s_2 \geq P - b_3 - b_1;$ $s_3 \geq P - b_2 - b_1$	$(2^3), (2), (2^4)$
E, $(1, 2, 2, 3, 3; -1, -1)$	$s_4 \geq a_4; s_5 \geq a_5;$ $s_1 + s_2 + s_6 - s_7 - s_8 \geq a_2 + \dots + a_5;$ $s_3 \geq P - b_2 - b_1$	$(1^3), (1, 1), (1^5)$
F, $(2, 3, 1, 2, 3; -2, -1)$	$s_2 \geq a_2; s_5 \geq a_5; s_1 + s_4 - s_8 \geq$ $a_1 + a_2 + a_4 + a_5 - b_1;$ $s_3 \geq P - b_2 - b_1$	$(1^4), (2), (3, 1, 1, 1)$

The one-box tiles such as $s_4 + s_6 - s_8 \geq a_4 + a_5 - b_2$ ($I = (4, 5, 6, 7), J = (1, 2, 3, 5), K = (4, 5, 6, 8)$) are the $(m, 3)$ versions of the Pieri triples $(\text{Pi})_i$ of [Mtwo] Lemma 4.3; the three-index tiles $s_1 + s_5 + s_6 - s_7 - s_8 \geq a_3 + a_4 + a_5$ and $s_1 + s_2 + s_6 - s_7 - s_8 \geq a_2 + \dots + a_5$ add two boxes. The extra form at $z = 0$, $m = 4$, has the certificate $s_4 \geq a_4; s_1 \geq a_3 + a_4$ ($I = J = (1, 4), K = (3, 4), \lambda = (2), (2), (2, 2)$); $s_2 + s_5 - s_6 \geq a_2 + a_3 + a_4 - b_1$ ($I = (1, 2, 4, 5), J = (1, 2, 4, 7), K = (2, 3, 4, 7), \lambda(I) = (1, 1), \lambda(J) = (3, 1), \lambda(K) = (3, 1, 1, 1)$); $s_3 \geq P - b_2 - b_1$ — the $(3, 1)$ tile is the one that does not survive padding (its instantiation at $z' = 1$ has no realisation with $LR \geq 1$, in the reviewer's dimension-free search), consistent with Theorem 3.

Dimension-free evidence: the tile signatures $(K_a, K_b, U, N_{\text{off}})$ (with N measured from d) coincide between $z = 1, 2, 3$ for every form when $m \leq 6$ (LP duals are not unique, so the few differences at $m = 4, 7$ are choices, not obstructions); and every tile of every $z = 1$ certificate for $m = 3, 4, 5$ ($30 + 64 + 106$ tiles) is realisable with $\text{LR} \geq 1$ at $z' = 2, 3, 4$ (reviewer's `tiles_dimfree2.py`). This supports, but does not prove, the general- z validity of the B-F certificates.

5. Upper bounds

5.1 3×3 -block weighted shifts

As in [Mtwo] Section 5.1: choose a layering $\Lambda_0, \dots, \Lambda_T$ of the nonzero spectrum together with some zeros (layers of size ≤ 3 here), let $D_t = \text{diag } \Lambda_t$, and take $C = \sum_{t=1}^T M_t$ block-bidiagonal with $M_t : \Lambda_{t-1} \rightarrow \Lambda_t$. Then $CC^* = \bigoplus_t 2R_t$ (R_t on Λ_t), $C^*C = \bigoplus_t 2S_t$ (S_t on Λ_{t-1}), and $CC^* - C^*C = 2F$ iff

$$S_1 = -D_0, \quad R_t - S_{t+1} = D_t \quad (1 \leq t < T), \quad R_T = D_T, \quad R_t \text{ and } S_t \text{ have the same nonzero spectrum.} \quad (5.1)$$

Given PSD R_t, S_t with equal nonzero spectra, $M_t = V\sqrt{2\Sigma}W^*$ realises them; zeros of F left outside the layers lie in $\ker C \cap \ker C^*$. The cost is $\frac{1}{2}\|C\|^2 = \sum_t \text{Tr } S_t = \sum_{t=1}^T \text{tail}_t = \text{cost}(L)$ (inserted zeros change nothing), and $\text{rank } C = \sum_t \text{rank } S_t \leq \sum_t \min(|\Lambda_{t-1}|, |\Lambda_t|)$. A layering is *feasible* at λ if (5.1) has a solution; then $\kappa_d(\lambda) \leq \text{cost}(L)(\lambda)$.

5.2 Chain feasibility as a linear program, and its convexity

Fix a *symbolic* layering (layers are lists of symbols $b_i, a_j, 0$), and let $n_t = |\Lambda_t| \leq 3$. For a Hermitian $n \times n$ sum $\gamma = \text{spec}(\alpha + \beta)$ the Horn conditions are the trace identity and the inequalities indexed by T_r^n , $1 \leq r < n$: for $n = 2$ the three Weyl inequalities, for $n = 3$ the twelve inequalities of $T_1^3 \cup T_2^3$ (six Weyl and six Ky Fan–Lidskii type). These are necessary (Weyl, Lidskii, Horn) and sufficient for every n (Klyachko [Kl98], Knutson–Tao [KT99]; see Fulton [Fu00]); here only $n \leq 3$ is needed. Consider the linear program in the variables $\sigma^{(t)} \in \mathbb{R}^3$, $t = 1, \dots, T$:

$$\sigma_1^{(t)} \geq \sigma_2^{(t)} \geq \sigma_3^{(t)} \geq 0, \quad \sigma_k^{(t)} = 0 \text{ for } k > \min(n_{t-1}, n_t), \quad \sigma^{(1)} = (-\Lambda_0)^\downarrow, \quad \sigma^{(T)} = \Lambda_T^\downarrow, \\ (\sigma^{(t)}|_{n_t}, (-\Lambda_t)^\downarrow, \sigma^{(t+1)}|_{n_t}) \text{ satisfies the Horn conditions in dimension } n_t \quad (1 \leq t < T), \quad (5.2)$$

where $x^\downarrow$ is the decreasing rearrangement padded with zeros to length 3 and $|_n$ the first n entries.

Lemma 5.1 (chain feasibility is the LP). The layering is feasible at λ iff (5.2) is feasible, and then $\sigma^{(t)}$ is the spectrum of S_t (and of R_t).

Proof. If (5.1) has a solution, $\sigma^{(t)} := \text{spec } S_t$ satisfies (5.2): the rank cap because S_t lives on Λ_{t-1} and R_t , with the same nonzero spectrum, on Λ_t ; the step conditions because $S_{t+1} = R_t + (-D_t)$ is a Hermitian sum on Λ_t (Horn necessity). Conversely let σ satisfy (5.2). By Lemma 5.0 of [Mtwo] — feasibility depends on the past only through $\text{spec } S_t$, since R_t may be *any* PSD matrix on Λ_t with the nonzero spectrum of S_t — construct inductively: $S_1 = -D_0$; given S_t with spectrum $\sigma^{(t)}$, Horn sufficiency in dimension n_t provides a PSD R_t on Λ_t with spectrum $\sigma^{(t)}|_{n_t}$ such that $R_t - D_t$ has spectrum $\sigma^{(t+1)}|_{n_t} \geq 0$; set $S_{t+1} = R_t - D_t$. At the end $\sigma^{(T)} = \Lambda_T^\downarrow$ lets us take $R_T = D_T$. $\square$

Lemma 5.2 (convexity). For a fixed symbolic layering, the set of λ in the closed stratum at which it is feasible is convex.

Proof. On the closed stratum (a and b ordered, all entries ≥ 0) the decreasing rearrangement of a layer is a *fixed linear* function of λ : within $-\Lambda_t$ the b 's come first in index order, then the zeros, then the $-a_j$ in reverse index order, and this order never changes; likewise $(-\Lambda_0)^\downarrow$ and $\Lambda_T^\downarrow$. Hence every constraint of (5.2) is linear in (λ, σ) jointly, the feasible set is a polyhedron in (λ, σ) -space, and the set of feasible λ is its projection, which is convex. $\square$

The isospectral condition in (5.1) is not convex in matrix space; the argument never uses matrix space, only spectra. The encoding (5.2) is `layering_lp.feasible` (author, HiGHS) and `chain_own.chain_feasible_exact` (reviewer, exact rational LP over the reviewer's own T_r^n , $n \leq 3$); the reviewer checked the author's encoding line by line (rank caps, ordering, trace, terminal condition).

Corollary 5.3. If $g \in S(m, z)$ and one symbolic layering with cost $= g$ is feasible at every vertex of the polytope C_g , then $\kappa_d \leq g$ on C_g ; combined with $\kappa_d \geq \max S$ this gives $\kappa_d = g$ on C_g , and since the chambers cover the stratum, $\kappa_d = \max S$ everywhere.

5.3 Chamber-vertex verification

For every $g \in S(m, z)$ the vertices of C_g were enumerated — exact brute force over constraint subsets for $m \leq 4$ and `qhull HalfspaceIntersection` for $m \geq 5$ (author), exact rational enumeration with `pycddlib/GMP` for $m \leq 8$ (reviewer), with identical counts — and the level layering $L(g)$, or $L(g)$ with one zero inserted, was tested at each vertex (`coverage_vertices.py`; reviewer `chain_own.py`). Vertex counts per chamber set:

m	3	4	5	6	7	8	9	total
$z = 0$	50	122	218	320	510	756	969	2945
$z = 1$ and $z = 2$	50	117	212	320	501	746	969	2915 each

In total 787 chambers ($265 + 261 + 261$) and 8 775 vertices. In 779 chambers the base layering $L(g)$ is feasible at every vertex; in the remaining eight it is not, but one layering with a single zero inserted is (Section 5.4). Runtimes: ≤ 27 s per (m, z) for the author's float run; seconds for $m \leq 7$, 85–112 s per (m, z) at $m = 8$ and 542–546 s per (m, z) at $m = 9$ for the exact run (the $m = 9$ rows were pending when the review was written and were completed for this manuscript with the reviewer's script, `chain_m9.py`; all 3×969 vertices exact, every chamber covered, no zero insertion at $m = 9$).

5.4 The eight chambers with an inserted zero

They are the chambers, at $z = 1$ and at $z = 2$, of

$$\begin{aligned}
 &(1, 1, 2, 3; -1, 0) [m = 4, D], & (1, 2, 2, 3, 3; -1, -1) [m = 5, E], \\
 &(1, 1, 2, 2, 2, 3, 4; -1, 0) [m = 7, D], & (1, 2, 2, 2, 3, 3, 4, 4; -1, -1) [m = 8, E],
 \end{aligned}$$

with feasible chains

$$\begin{aligned}
 &\{b_2, b_3\}\{b_1, a_1, a_2\}\{a_3, 0\}\{a_4\}, & \{b_3\}\{b_1, b_2, a_1\}\{a_2, a_3, 0\}\{a_4, a_5\}, \\
 &\{b_2, b_3\}\{b_1, a_1, a_2\}\{a_3, a_4, a_5\}\{a_6, 0\}\{a_7\}, & \{b_3\}\{b_1, b_2, a_1\}\{a_2, a_3, a_4\}\{a_5, a_6, 0\}\{a_7, a_8\} :
 \end{aligned}$$

the zero goes into the penultimate layer. These are exactly the $m \equiv 1, 2 \pmod{3}$ cases where $z = 0$ has the extra form $g_0^{(m)}$; at $z = 0$ every base chain is feasible on its (smaller) chamber. After padding

$g_0^{(m)}$ is no longer valid, the neighbouring chamber grows, and on the new part the base chain — whose rank is at most m by the bound of Section 5.1 (e.g. $2 + 1 + 1$ for the $m = 4$ chain) — fails, while the chain with a zero has rank at most $m + 1$. This is the $(m, 3)$ analogue of Theorem C'(3) of [Mtwo], and it matches the exact rank forcing of Theorem 3(3).

Explicit matrices. `construct_C.py` turns the argmax layering at a point into a chain by the LP (5.2) with maximal Horn slack, solves each step (a 2×2 step in closed form, a 3×3 step with a rank-one target by the secular equation, otherwise by least squares on the characteristic polynomial polished on the eigenvalues), and assembles $C = \oplus M_t$ (real, block bidiagonal). At 500 points per (m, z) , $m = 3, \dots, 9$, $z = 0, 1$ (7000 matrices; a quarter from a tie-heavy law, every eighth point with an exact tie), all points were constructed; $\max |CC^\top - C^\top C - 2F|$ has median $\leq 3 \cdot 10^{-15}$ and exceeds 10^{-9} at 5 points ($\leq 3.0 \cdot 10^{-6}$, near-degenerate steps); $|\frac{1}{2}\|C\|^2 - \max S| \leq 8 \cdot 10^{-14}$ except at those points. The zero-inserted chain was needed at 45/10/6/3 of 500 points for $m = 4/5/7/8$ at $z = 1$ and never at $z = 0$ or for $m = 3, 6, 9$. $\text{rank } C = m$ everywhere else, except that the maximal-slack interior chain has rank $m + 1$ at 52–58 of 500 points for $m = 3$ and at 2–14 for $m = 5$, where rank- m optimizers nevertheless exist (Section 6). The reviewer's own constructor reproduces this at 28 points ($m \leq 7$).

6. Padding and rank: proof of Theorem 3

Sets. $S(m, z)$ for $z = 0, 1, 2$ (and $z = 3, m \leq 6$) were computed by the closure loop of Section 8; comparing the files gives (1) and the identity $S(m, 0) = S(m, 1) \cup \{g_0^{(m)}\}$ in (2), the sets $S(m, 1), S(m, 2)$ (and $S(m, 3)$) being identical. By Theorem 1, $\kappa_{m+3+z} = \max S(m, z)$, which gives the padding invariance statements.

Strict decrease, exactly. For $m = 4, 5, 7, 8$ take the rational point (rounded strict Chebyshev centre of the chamber of $g_0^{(m)}$ in $S(m, 0)$, renormalised to $P = 1$; `exact_strict.py`):

$$m = 4 : a = (57/200, 157/600, 143/600, 43/200), b = (107/300, 1/3, 31/100);$$

$$m = 5 : a = (7/30, 13/60, 1/5, 11/60, 1/6), b = (7/20, 1/3, 19/60);$$

$$m = 7 : a = (1/6, 19/120, 3/20, 43/300, 27/200, 19/150, 3/25), b = (41/120, 1/3, 13/40);$$

$$m = 8 : a = (11/75, 17/120, 27/200, 77/600, 73/600, 23/200, 13/120, 31/300), b = (17/50, 1/3, 49/150).$$

There $g_0^{(m)} = 401/300, 29/20, 41/24, 277/150$ and $\max S(m, 1) = 787/600, 43/30, 17/10, 46/25$, margins $1/40, 1/60, 1/120, 1/150$. Now $\kappa_{m+3} \geq g_0^{(m)}$ by its exact tiling certificate ($d = m + 3 \leq 11$), and $\kappa_{m+4} = \max S(m, 1)$ exactly ($d = m + 4 \leq 12$, exact lower bound; exact upper bound by Theorem 1). Hence $\kappa_{m+3}(\lambda) > \kappa_{m+4}(\lambda \oplus 0)$ at these points, and by continuity of both sides on an open set. Since $\kappa_{m+3+z}(\lambda \oplus 0^z) \leq \kappa_{m+4}(\lambda \oplus 0) < g_0^{(m)}(\lambda)$ for every $z \geq 1$, $g_0^{(m)}$ is invalid in every padded dimension. At the padding example $a = (17, 17, 17, 10)/61, b = (25, 18, 18)/61$ one has $g_0^{(4)} = 74/61$ and $\max S(4, 1) = 73/61$.

Rank, exactly. By Horn sufficiency in dimension d , every feasible s of (2.1) is the common spectrum of some C , and every optimizer has $s_d = 0$ (otherwise shifting lowers the cost), so the set of optimal common spectra is the optimal face of (2.1), and $\text{rank } C = \#\{t : s_t > 0\}$. The reviewer's GMP linear programs over the full Fulton lists (2062 triples for $d = 7$, 8752 for $d = 8$) at the integer spectrum $(17, 17, 17, 10, [0], -18, -18, -25)$ give: $\kappa_7 \cdot 61 = 74$ with an optimal $s = (27, 18, 18, 10, 1, 0)$ and $\min s_5$ on the optimal face $= 0$, so a rank-4 optimizer exists (rank ≥ 4 is the inertia obstruction, [RankOnset] Lemma 2.1); $\kappa_8 \cdot 61 = 73$ with $s = (26, 18, 18, 10, 1, 0, 0)$, $\min s_5$ on the optimal face $= 1$,

so every optimizer has rank ≥ 5 ; and the minimum of (2.1) with $s_5 = s_6 = s_7 = 0$ is 74. This is (3). The author’s float scan (`rank_scan.py`, 150 random points per (m, z) , $m \leq 7$, $z \leq 1$) finds minimal optimal rank m at every random point except one $m = 4, z = 1$ point of rank 5, and the tail-maximising optimizer of rank $m + 1$ at 2–35% of points; the reviewer’s sampling of the region $\{g_0^{(m)} > \max S(m, 1)\}$ gives (4).

Beyond $m \leq 9$ (numerical). At $z = 0$ the predicted set of Conjecture 4 is exceeded at 4/1200 points for $m = 10$ and 1/1200 for $m = 11$ (by up to $2.4 \cdot 10^{-2}$), each time with the single dual form listed in (4); the reviewer confirmed each to be valid at $z = 0$ (joint LP, $\min(\kappa - g) \geq -3.5 \cdot 10^{-15}$), invalid at $z = 1$ ($-0.0333, -0.0303$) and exposed relative to the predicted set (chamber radius $3.9 \cdot 10^{-3}, 3.4 \cdot 10^{-3}$), found the $m = 13$ continuation (valid at $z = 0$, invalid at $z = 1$ by 0.0256, radius $2.3 \cdot 10^{-3}$), and no extra form at $m = 12$ in 600 samples. The layerings of these forms all have schedule $(2, 1, 0)$ with the a ’s in order, $\{b_3\}\{b_2, a_1\}\{b_1, a_2, a_3\}$ followed by triples and a shrinking tail; a rule “extra form iff 3 does not divide m ” is observed up to $m = 13$ and not proved.

7. Consistency checks

Reduction to inertia $(m, 2)$. Restricting the forms of $S(m, z \geq 1)$ to $b_3 = 0$ must reproduce $\Phi = \max(F_k, G_j)$ of [Mtwo] Theorem A, since κ is continuous. Exactly (reviewer, `m2_reduction.py`, $m = 4, 5$): every $(m, 2)$ form $F_0, \dots, F_{m-1}, G_1, G_3, \dots$, reduced modulo $\sum a = b_1 + b_2$, belongs to $S(m)|_{b_3=0}$, and on each $(m, 2)$ chamber the exact maximum of $h - \varphi$ over $h \in S(m)$ is 0 (GMP LPs), so $\max S|_{b_3=0} = \Phi$ identically. Numerically (author), $\max S(m, z \geq 1)|_{b_3=0} = \Phi$ to $2 \cdot 10^{-15}$ at 3000 random points per $m \leq 9$. The family-A forms restrict to the F_k .

Reduction to one spike. At $b_2 = b_3 = 0$ the value is $\sum_j j a_j$ ([OneSpike] Theorem 3.1); this form belongs to $S(m)$ and dominates every other form on that face (exact for $m = 4, 5$; $4 \cdot 10^{-15}$ at random points for $m \leq 9$).

Rank-adaptive bound. $P \leq \kappa_d$ ([RankAdaptive] Theorem 1.1) is the form $(1, \dots, 1; 0, 0)$, the cost of the family-A layering $\{b_1, b_2, b_3\}\{a_1, a_2, a_3\}\{a_4, a_5, a_6\} \dots$, exposed for $m \geq 4$.

Fresh samples. Independently of the chamber argument, κ_d (hive LP, [KT99] honeycomb encoding, calibrated against the explicit Horn LP for $d \leq 9$) was compared with $\max S(m, z)$ at 2000 fresh points per (m, z) for $m = 3, \dots, 9$, $z \in \{0, 1, 3\}$ ($d \leq 14$), seven sampling laws including tie-heavy, dominant- a_1 , flat- b and the neighbourhood of the padding example: with default HiGHS tolerances the signed gap $\kappa - \max S$ is $\leq 6.2 \cdot 10^{-15}$ (never a missing form) while $\max S - \kappa$ reaches $2.5 \cdot 10^{-7}$ at a few near-tie points; every such point re-solved with feasibility tolerances 10^{-10} gives $|\kappa - \max S| \leq 2.2 \cdot 10^{-15}$. For $(9, 2)$ the original run used an older script without the re-solve and left a residual of $1.8 \cdot 10^{-7}$; the rerun with the re-solve procedure for this manuscript (`verify_formula2.py` 9 9 2000 2 2026, 13 s) re-solves 17 points and leaves $|\kappa - \max S| \leq 1.3 \cdot 10^{-15}$, and the reviewer’s independent tight run gives $5.3 \cdot 10^{-15}$. The reviewer’s own hive LP at 250 points for each of the 21 pairs $m = 3, \dots, 9$, $z \leq 2$ agrees with $\max S$ to $6 \cdot 10^{-15}$, and his explicit Horn LP ($d \leq 9$) to $1.1 \cdot 10^{-15}$.

8. Verification

Discovery and closure (author). Hive-LP duals at 1500 + 9000 random points per (m, z) , $m \leq 7$, $z \leq 4$, rationalised, then filtered by the validity LP and the exposedness (chamber-radius) LP (`m3_forms.py`, `m3_gather.py`, `m3_analyze.py`, `m3_tools.py`); then the closure loop `m3_closure.py` (validity + exposedness filter, chamber vertices, hive LP at every vertex, new dual forms at any gap, repeat) for

$m = 2, \dots, 9, z \leq 2$ ($z \leq 3$ for $m \leq 6$): every set closes in one iteration with maximal vertex gap $\leq 1.8 \cdot 10^{-14}$ and no new form (closure_all.log, closure_z2.log). The vertex-gap check is the same argument as Corollary 5.3 with $\kappa_d - g$ convex in place of a chain, and is what proves *completeness* of S for (8, 2), (9, 1), (9, 2), where no exact certificates exist. Horn-LP duals over the full Fulton lists (522, 2062, 8752, 39716 triples for $d = 6, \dots, 9$) confirmed the tiling structure of the certificates before the catalogue was built (m3_horn_duals.py, duals_m*_z*.log).

Certificates (author). certs.py m, z for all $d \leq 12$ (certs_all.log, ≤ 14 s each): 10/10, ..., 69/69 forms certified; tile counts as in Section 4.1. familyA_cert.py 12 4 (2 s): 945 exact family-A certificates.

Coverage (author). coverage_vertices.py m, z for $m \leq 9, z \leq 2$ (coverage_all.log, coverage_m*_z*.json): 787 chambers, 8775 vertices, every chamber covered by a single chain feasible at all its vertices; the eight zero-insertions of Section 5.4. Matrices: construct_C.py (Section 5.4; 3–175 s per (m, z) ; construct_all.log contains one duplicated $m = 8$ line and one superseded $m = 5, z = 0$ line from an earlier solver version, which are not counted). Rank: rank_scan.py (rank_all.log; the entries “(7,6)” there are a threshold artefact). Families: families2.py reproduces $S(m, 1)$ for $m \leq 9$; test_pred.py at $m = 10, z = 1$: 1200 points, $|\kappa - \max \text{pred}| \leq 6.2 \cdot 10^{-15}$ with the predicted 87 forms.

Independent review (reviewer, own code in repro/reviewer/). Tools written from scratch and calibrated first: Fulton recursion for T_r^n (3, 12, 41, 142, 522, 2062, 8752, 39716 triples for $n = 2, \dots, 9$, same counts as the author); LR coefficients by Jacobi-Trudi + Pieri, agreeing with T_r^n membership on all 5521 sum-condition triples with $n \leq 7$ and with lr.py, lr2.py on 1500 random triples; Horn LP and hive LP in an own encoding agreeing to 10^{-14} ; exact rational LP and vertex enumeration (pycddlib); known values $\kappa_4(3, -1, -1, -1) = 6$, $\kappa_5(4, -1^4) = 10$, $\kappa_7 = 74/7$, $\kappa_8 = 73/7$ reproduced. Results: (a) tiles_check.py: 36 certificates, 149 tiles, all verified; dimension-free instantiation of tile signatures realisable at $z' = 3$ (24/24), $z' = 1$ (79/82, the three failures being the special tiles of the $z = 0$ -only forms) and $z' = 0$ (102/104, two $z \geq 1$ tiles with $U \ni m + 1$ colliding with $N \ni d - 2$ at $d' = m + 3$, i.e. not meant for $z = 0$); tiles_dimfree2.py as in Section 4.3. (b) chain_own.py: exact vertices and exact chain LP, $m = 3, \dots, 8, z \leq 2$ — identical vertex counts, identical eight zero-insertions. (c) formula_check.py, extra_checks.py: Section 7. (d) padding_check.py: Theorem 3(3) exactly and the rank forcing on the new part for $m = 4$; extra_checks.py for $m = 5, 7$. (e) exposed_m5.py: all 27 ($z = 0$) and 26 ($z = 1$) forms at $m = 5$ have chamber radius $\geq 1.7 \cdot 10^{-2}$ with $\kappa = g$ at the centre, and each is the unique argmax at least once in 6000 samples. (f) pred_m11.py, complete_m11.py, extra_checks.py, z0_extra.py: Conjecture 4 at $m = 10, 11, 12$ (200–300 points each, $|\kappa - \max \text{pred}| \leq 7 \cdot 10^{-15}$); at $m = 11$ all 103 forms valid ($\min(\kappa - g) \geq -4.6 \cdot 10^{-15}$) and exposed (radius $\geq 5.65 \cdot 10^{-3}$); chamber-vertex completeness at $m = 10$ (87 chambers, 1353 vertices, gap $\leq 3.0 \cdot 10^{-14}$) and $m = 11$ (103 chambers, 1781 vertices, $\leq 4.6 \cdot 10^{-14}$); the $z = 0$ extras of Theorem 3(4). (g) m2_reduction.py: Section 7. (h) construct_own.py: 28 explicit matrices. The reviewer’s reporting corrections (exact certificates only for $d \leq 12$; vertex total 8775; the (9, 2) residual; the convexity argument via Horn sufficiency and Lemma 5.0; the family-A docstring; rank evidence for $m = 5, 7$; “multipliers rounded to 1 and verified exactly”) are applied throughout.

Reruns for this manuscript (repro/run_fast.sh, 2026-09-13, one core; logs and timings in repro/logs/): verify_formula2.py 9 9 2000 2 2026 (14 s), certs.py 4 1 (1 s), familyA_cert.py 12 4 (1 s), families2.py 3 9, coverage_vertices.py 4 1 (23 s), test_pred.py 10 1 300 (3 s); reviewer’s tiles_check.py, m2_reduction.py, padding_check.py (8 s), chain_own.py 5 (3 s), pred_m11.py (7 s), complete_m11.py (13 s), exposed_m5.py (4 s); and exact_strict.py (Section 6). All outputs agree with the quoted ones. Reviewer’s exact chain check at $m = 9$ (chain_m9.py

z , 542, 544, 546 s for $z = 0, 1, 2$; logs/chain_own_m9_z{z}.log): 969 exact vertices each, every chamber covered by its base chain.

9. Open questions

1. *A general- m upper bound.* The 3×3 chain has no one-parameter recursion like Lemma 5.2 of [Mtwo]; the chain LP (5.2) at chamber vertices is the substitute. A closed description of the chain-feasibility regions of the families A–F would prove Conjecture 4’s upper bound.
2. *General- z validity of the B–F tiles* by the Pieri strip condition per tile family (Section 4.3).
3. *Exhaustiveness of the six families* and the origin of the quadratic count; a formula for $|S(m)|$.
4. *The padding rule.* Prove that an extra exposed form exists at $z = 0$ iff 3 does not divide m , identify its layering for all m , and prove the rank- $(m + 1)$ forcing on the new part in general.
5. *Inertia (m, n) , $n \geq 4$.* Whether $n \times n$ -block shifts attain and tiles with $\lambda(J)$ of at most n boxes certify; the $d = 8$ inertia- $(4, 4)$ phase of [RankOnset] Theorem 4.1 shows that rank $\max(n_+, n_-)$ cannot persist.
6. *Optimizers.* A classification of chambers by minimal optimal rank and by uniqueness of the optimal spectrum (only sampled here).

10. Reproducibility

The directory repro/ accompanying this manuscript contains: author/ — every script, JSON file, pickle and log of the author’s working directory (cycle2/work/A3_inertia_m3/), including closed_m{m}_z{z}.json (the sets $S(m, z)$), certs_m{m}_z{z}.json (every tiling certificate), coverage_m{m}_z{z}.json, pred_m{m}.json ($m \leq 14$) and the logs quoted above; reviewer/ — the reviewer’s scripts and logs, with the only modification that two absolute paths were replaced by paths relative to repro/ (marked in the source), plus chain_m9.py, a wrapper running the reviewer’s exact chain check at $m = 9$ one z per process; exact_strict.py (Section 6); run_fast.sh and logs/ with the reruns and timings; and README.md with commands, runtimes and expected final lines. Dependencies: Python 3.12, numpy, scipy (HiGHS, qhull), pycddlib (GMP) for the exact reviewer scripts. Literature queries (2026-09-13, three WebSearch queries on the Hilbert–Schmidt inverse self-commutator cost with prescribed spectrum, on block weighted shifts with prescribed self-commutator spectrum and Pieri certificates, and on follow-ups to Angel–Schechtman) returned only commutator norm inequalities, almost-commuting matrices, limiting Horn inequalities (arXiv:2410.08907), weighted-shift spectral theory, and [JOS13, AS15].

AI-assistance statement

The mathematics, computations and text of this manuscript were produced by Claude Fable 5.1 agents (Anthropic) working under the direction of the author, who set the problem, selected the line of records and is responsible for the claims. A first agent performed the discovery computations and was interrupted by a usage limit; a second agent resumed from its files and completed the certificates, the coverage and the families; a third, independent agent carried out an adversarial review with its own implementations of every ingredient (Fulton recursion, Littlewood–Richardson coefficients, Horn and hive LPs, exact rational LPs and vertex enumeration), upgraded the upper bound to exact arithmetic for $m \leq 8$ (its script was rerun at $m = 9$ for this manuscript), strengthened the conjecture at $m = 10, 11$ and listed the reporting corrections applied here; a fourth agent wrote this text, added the exact strict-

decrease points of Section 6, ran the reviewer's exact check at $m = 9$ and reran the fast scripts. No fall-back to another model occurred in any of these sessions. The manuscript has not been peer reviewed by humans.

Appendix A. The exposed forms for $m \leq 6$

Forms in canonical coordinates $(\alpha_1, \dots, \alpha_m; \beta_1, \beta_2)$, with family, b -schedule and level layering; the sets are $S(m, z \geq 1)$, and the $z = 0$ -only forms are appended. Larger m : repro/author/closed_m{m}_z{z}.json.

$m = 3$ (10 forms):

form	fam.	u	layering
$(-1, 0, 1; 1, 2)$	C	$(1, 0, 2)$	$\{b_2\}\{b_1, a_1\}\{b_3, a_2\}\{a_3\}$
$(0, 0, 1; 1, 1)$	A	$(0, 0, 1)$	$\{b_1, b_2\}\{b_3, a_1, a_2\}\{a_3\}$
$(0, 1, 1; 1, 0)$	A	$(0, 1, 1)$	$\{b_1\}\{b_2, b_3, a_1\}\{a_2, a_3\}$
$(0, 1, 2; 1, -1)$	B	$(0, 2, 1)$	$\{b_1\}\{b_3, a_1\}\{b_2, a_2\}\{a_3\}$
$(1, 0, 1; 0, 1)$	C	$(1, 0, 1)$	$\{b_2\}\{b_1, b_3, a_2\}\{a_1, a_3\}$
$(1, 2, 1; 0, -1)$	B	$(0, 1, 0)$	$\{b_1, b_3\}\{b_2, a_1, a_3\}\{a_2\}$
$(2, 1, 1; -1, 0)$	D	$(1, 0, 0)$	$\{b_2, b_3\}\{b_1, a_2, a_3\}\{a_1\}$
$(2, 2, 1; -1, -1)$	E	$(1, 1, 0)$	$\{b_3\}\{b_1, b_2, a_3\}\{a_1, a_2\}$
$(2, 3, 1; -2, -1)$	F	$(2, 1, 0)$	$\{b_3\}\{b_2, a_3\}\{b_1, a_1\}\{a_2\}$
$(3, 1, 2; -2, -1)$	F	$(2, 1, 0)$	$\{b_3\}\{b_2, a_2\}\{b_1, a_3\}\{a_1\}$

$m = 4$ (18 forms, plus one at $z = 0$):

form	fam.	u	layering
$(-1, 0, 0, 1; 2, 1)$	A	$(0, 1, 2)$	$\{b_1\}\{b_2, a_1\}\{b_3, a_2, a_3\}\{a_4\}$
$(-1, 0, 1, 0; 1, 2)$	C	$(1, 0, 2)$	$\{b_2\}\{b_1, a_1\}\{b_3, a_2, a_4\}\{a_3\}$
$(-1, 0, 1, 1; 2, 0)$	A	$(0, 2, 2)$	$\{b_1\}\{a_1\}\{b_2, b_3, a_2\}\{a_3, a_4\}$
$(-1, 0, 1, 2; 2, -1)$	B	$(0, 3, 2)$	$\{b_1\}\{a_1\}\{b_3, a_2\}\{b_2, a_3\}\{a_4\}$
$(0, -1, 0, 1; 1, 2)$	C	$(1, 0, 2)$	$\{b_2\}\{b_1, a_2\}\{b_3, a_1, a_3\}\{a_4\}$
$(0, 0, 1, 1; 1, 1)$	A	$(0, 0, 1)$	$\{b_1, b_2\}\{b_3, a_1, a_2\}\{a_3, a_4\}$
$(0, 1, 1, 1; 1, 0)$	A	$(0, 1, 1)$	$\{b_1\}\{b_2, b_3, a_1\}\{a_2, a_3, a_4\}$
$(0, 1, 1, 2; 0, 1)$	C	$(1, 0, 1)$	$\{b_2\}\{b_1, b_3, a_1\}\{a_2, a_3\}\{a_4\}$
$(0, 1, 2, 1; 1, -1)$	B	$(0, 2, 1)$	$\{b_1\}\{b_3, a_1\}\{b_2, a_2, a_4\}\{a_3\}$
$(1, 0, 1, 1; 0, 1)$	C	$(1, 0, 1)$	$\{b_2\}\{b_1, b_3, a_2\}\{a_1, a_3, a_4\}$
$(1, 1, 1, 2; 0, 0)$	A	$(0, 0, 0)$	$\{b_1, b_2, b_3\}\{a_1, a_2, a_3\}\{a_4\}$
$(1, 1, 2, 3; -1, 0)$	D	$(1, 0, 0)$	$\{b_2, b_3\}\{b_1, a_1, a_2\}\{a_3\}\{a_4\}$
$(1, 2, 1, 2; 0, -1)$	B	$(0, 1, 0)$	$\{b_1, b_3\}\{b_2, a_1, a_3\}\{a_2, a_4\}$
$(2, 1, 1, 2; -1, 0)$	D	$(1, 0, 0)$	$\{b_2, b_3\}\{b_1, a_2, a_3\}\{a_1, a_4\}$
$(2, 1, 2, 3; -1, -1)$	E	$(1, 1, 0)$	$\{b_3\}\{b_1, b_2, a_2\}\{a_1, a_3\}\{a_4\}$
$(2, 2, 1, 2; -1, -1)$	E	$(1, 1, 0)$	$\{b_3\}\{b_1, b_2, a_3\}\{a_1, a_2, a_4\}$
$(2, 3, 1, 2; -2, -1)$	F	$(2, 1, 0)$	$\{b_3\}\{b_2, a_3\}\{b_1, a_1, a_4\}\{a_2\}$
$(3, 1, 2, 2; -2, -1)$	F	$(2, 1, 0)$	$\{b_3\}\{b_2, a_2\}\{b_1, a_3, a_4\}\{a_1\}$
$(1, 2, 3, 4; -2, -1)$	$z = 0$ only	$(2, 1, 0)$	$\{b_3\}\{b_2, a_1\}\{b_1, a_2\}\{a_3\}\{a_4\}$

$m = 5$ (26 forms, plus one at $z = 0$):

form	fam.	u	layering
$(-2, -1, -1, 0, 1; 2, 3)$	C	$(1, 0, 3)$	$\{b_2\}\{b_1, a_1\}\{a_2, a_3\}\{b_3, a_4\}\{a_5\}$
$(-2, -1, 0, 0, 1; 3, 1)$	A	$(0, 2, 3)$	$\{b_1\}\{a_1\}\{b_2, a_2\}\{b_3, a_3, a_4\}\{a_5\}$
$(-2, -1, 0, 1, 1; 3, 0)$	A	$(0, 3, 3)$	$\{b_1\}\{a_1\}\{a_2\}\{b_2, b_3, a_3\}\{a_4, a_5\}$
$(-2, -1, 0, 1, 2; 3, -1)$	B	$(0, 4, 3)$	$\{b_1\}\{a_1\}\{a_2\}\{b_3, a_3\}\{b_2, a_4\}\{a_5\}$
$(-1, -1, 0, 0, 1; 2, 2)$	A	$(0, 0, 2)$	$\{b_1, b_2\}\{a_1, a_2\}\{b_3, a_3, a_4\}\{a_5\}$
$(-1, 0, 0, 1, 1; 2, 1)$	A	$(0, 1, 2)$	$\{b_1\}\{b_2, a_1\}\{b_3, a_2, a_3\}\{a_4, a_5\}$
$(-1, 0, 1, 0, 1; 1, 2)$	C	$(1, 0, 2)$	$\{b_2\}\{b_1, a_1\}\{b_3, a_2, a_4\}\{a_3, a_5\}$
$(-1, 0, 1, 1, 1; 2, 0)$	A	$(0, 2, 2)$	$\{b_1\}\{a_1\}\{b_2, b_3, a_2\}\{a_3, a_4, a_5\}$
$(-1, 0, 1, 2, 1; 2, -1)$	B	$(0, 3, 2)$	$\{b_1\}\{a_1\}\{b_3, a_2\}\{b_2, a_3, a_5\}\{a_4\}$
$(0, -1, 0, 1, 1; 1, 2)$	C	$(1, 0, 2)$	$\{b_2\}\{b_1, a_2\}\{b_3, a_1, a_3\}\{a_4, a_5\}$
$(0, 0, 1, 1, 1; 1, 1)$	A	$(0, 0, 1)$	$\{b_1, b_2\}\{b_3, a_1, a_2\}\{a_3, a_4, a_5\}$
$(0, 1, 1, 1, 2; 1, 0)$	A	$(0, 1, 1)$	$\{b_1\}\{b_2, b_3, a_1\}\{a_2, a_3, a_4\}\{a_5\}$
$(0, 1, 1, 2, 1; 0, 1)$	C	$(1, 0, 1)$	$\{b_2\}\{b_1, b_3, a_1\}\{a_2, a_3, a_5\}\{a_4\}$
$(0, 1, 2, 1, 2; 1, -1)$	B	$(0, 2, 1)$	$\{b_1\}\{b_3, a_1\}\{b_2, a_2, a_4\}\{a_3, a_5\}$
$(1, 0, 1, 1, 2; 0, 1)$	C	$(1, 0, 1)$	$\{b_2\}\{b_1, b_3, a_2\}\{a_1, a_3, a_4\}\{a_5\}$
$(1, 1, 1, 2, 2; 0, 0)$	A	$(0, 0, 0)$	$\{b_1, b_2, b_3\}\{a_1, a_2, a_3\}\{a_4, a_5\}$
$(1, 1, 2, 2, 3; 0, -1)$	B	$(0, 1, 0)$	$\{b_1, b_3\}\{b_2, a_1, a_2\}\{a_3, a_4\}\{a_5\}$
$(1, 1, 2, 3, 2; -1, 0)$	D	$(1, 0, 0)$	$\{b_2, b_3\}\{b_1, a_1, a_2\}\{a_3, a_5\}\{a_4\}$
$(1, 2, 1, 2, 2; 0, -1)$	B	$(0, 1, 0)$	$\{b_1, b_3\}\{b_2, a_1, a_3\}\{a_2, a_4, a_5\}$
$(1, 2, 1, 2, 3; -1, 0)$	D	$(1, 0, 0)$	$\{b_2, b_3\}\{b_1, a_1, a_3\}\{a_2, a_4\}\{a_5\}$
$(1, 2, 2, 3, 3; -1, -1)$	E	$(1, 1, 0)$	$\{b_3\}\{b_1, b_2, a_1\}\{a_2, a_3\}\{a_4, a_5\}$
$(2, 1, 1, 2, 2; -1, 0)$	D	$(1, 0, 0)$	$\{b_2, b_3\}\{b_1, a_2, a_3\}\{a_1, a_4, a_5\}$
$(2, 1, 2, 3, 2; -1, -1)$	E	$(1, 1, 0)$	$\{b_3\}\{b_1, b_2, a_2\}\{a_1, a_3, a_5\}\{a_4\}$
$(2, 2, 1, 2, 3; -1, -1)$	E	$(1, 1, 0)$	$\{b_3\}\{b_1, b_2, a_3\}\{a_1, a_2, a_4\}\{a_5\}$
$(2, 3, 1, 2, 3; -2, -1)$	F	$(2, 1, 0)$	$\{b_3\}\{b_2, a_3\}\{b_1, a_1, a_4\}\{a_2, a_5\}$
$(3, 1, 2, 2, 3; -2, -1)$	F	$(2, 1, 0)$	$\{b_3\}\{b_2, a_2\}\{b_1, a_3, a_4\}\{a_1, a_5\}$
$(1, 2, 3, 3, 4; -2, -1)$	$z = 0$ only	$(2, 1, 0)$	$\{b_3\}\{b_2, a_1\}\{b_1, a_2\}\{a_3, a_4\}\{a_5\}$

$m = 6$ (34 forms, no extra at $z = 0$):

form	fam.	u	layering
$(-3, -2, -1, 0, 0, 1; 4, 1)$	A	$(0, 3, 4)$	$\{b_1\}\{a_1\}\{a_2\}\{b_2, a_3\}\{b_3, a_4, a_5\}\{a_6\}$
$(-3, -2, -1, 0, 1, 1; 4, 0)$	A	$(0, 4, 4)$	$\{b_1\}\{a_1\}\{a_2\}\{a_3\}\{b_2, b_3, a_4\}\{a_5, a_6\}$
$(-3, -2, -1, 0, 1, 2; 4, -1)$	B	$(0, 5, 4)$	$\{b_1\}\{a_1\}\{a_2\}\{a_3\}\{b_3, a_4\}\{b_2, a_5\}\{a_6\}$
$(-2, -1, -1, 0, 0, 1; 3, 2)$	A	$(0, 1, 3)$	$\{b_1\}\{b_2, a_1\}\{a_2, a_3\}\{b_3, a_4, a_5\}\{a_6\}$
$(-2, -1, -1, 0, 1, 0; 2, 3)$	C	$(1, 0, 3)$	$\{b_2\}\{b_1, a_1\}\{a_2, a_3\}\{b_3, a_4, a_6\}\{a_5\}$
$(-2, -1, 0, -1, 0, 1; 2, 3)$	C	$(1, 0, 3)$	$\{b_2\}\{b_1, a_1\}\{a_2, a_4\}\{b_3, a_3, a_5\}\{a_6\}$
$(-2, -1, 0, 0, 1, 1; 3, 1)$	A	$(0, 2, 3)$	$\{b_1\}\{a_1\}\{b_2, a_2\}\{b_3, a_3, a_4\}\{a_5, a_6\}$
$(-2, -1, 0, 1, 1, 1; 3, 0)$	A	$(0, 3, 3)$	$\{b_1\}\{a_1\}\{a_2\}\{b_2, b_3, a_3\}\{a_4, a_5, a_6\}$
$(-2, -1, 0, 1, 2, 1; 3, -1)$	B	$(0, 4, 3)$	$\{b_1\}\{a_1\}\{a_2\}\{b_3, a_3\}\{b_2, a_4, a_6\}\{a_5\}$
$(-1, -2, -1, 0, 0, 1; 2, 3)$	C	$(1, 0, 3)$	$\{b_2\}\{b_1, a_2\}\{a_1, a_3\}\{b_3, a_4, a_5\}\{a_6\}$
$(-1, -1, 0, 0, 1, 1; 2, 2)$	A	$(0, 0, 2)$	$\{b_1, b_2\}\{a_1, a_2\}\{b_3, a_3, a_4\}\{a_5, a_6\}$
$(-1, 0, 0, 1, 1, 1; 2, 1)$	A	$(0, 1, 2)$	$\{b_1\}\{b_2, a_1\}\{b_3, a_2, a_3\}\{a_4, a_5, a_6\}$

form	fam.	u	layering
$(-1, 0, 0, 1, 1, 2; 1, 2)$	C	$(1, 0, 2)$	$\{b_2\}\{b_1, a_1\}\{b_3, a_2, a_3\}\{a_4, a_5\}\{a_6\}$
$(-1, 0, 1, 0, 1, 1; 1, 2)$	C	$(1, 0, 2)$	$\{b_2\}\{b_1, a_1\}\{b_3, a_2, a_4\}\{a_3, a_5, a_6\}$
$(-1, 0, 1, 1, 1, 2; 2, 0)$	A	$(0, 2, 2)$	$\{b_1\}\{a_1\}\{b_2, b_3, a_2\}\{a_3, a_4, a_5\}\{a_6\}$
$(-1, 0, 1, 2, 1, 2; 2, -1)$	B	$(0, 3, 2)$	$\{b_1\}\{a_1\}\{b_3, a_2\}\{b_2, a_3, a_5\}\{a_4, a_6\}$
$(0, -1, 0, 1, 1, 1; 1, 2)$	C	$(1, 0, 2)$	$\{b_2\}\{b_1, a_2\}\{b_3, a_1, a_3\}\{a_4, a_5, a_6\}$
$(0, 0, 1, 1, 1, 2; 1, 1)$	A	$(0, 0, 1)$	$\{b_1, b_2\}\{b_3, a_1, a_2\}\{a_3, a_4, a_5\}\{a_6\}$
$(0, 1, 1, 1, 2, 2; 1, 0)$	A	$(0, 1, 1)$	$\{b_1\}\{b_2, b_3, a_1\}\{a_2, a_3, a_4\}\{a_5, a_6\}$
$(0, 1, 1, 2, 1, 2; 0, 1)$	C	$(1, 0, 1)$	$\{b_2\}\{b_1, b_3, a_1\}\{a_2, a_3, a_5\}\{a_4, a_6\}$
$(0, 1, 1, 2, 2, 3; 1, -1)$	B	$(0, 2, 1)$	$\{b_1\}\{b_3, a_1\}\{b_2, a_2, a_3\}\{a_4, a_5\}\{a_6\}$
$(0, 1, 2, 1, 2, 2; 1, -1)$	B	$(0, 2, 1)$	$\{b_1\}\{b_3, a_1\}\{b_2, a_2, a_4\}\{a_3, a_5, a_6\}$
$(1, 0, 1, 1, 2, 2; 0, 1)$	C	$(1, 0, 1)$	$\{b_2\}\{b_1, b_3, a_2\}\{a_1, a_3, a_4\}\{a_5, a_6\}$
$(1, 1, 1, 2, 2, 2; 0, 0)$	A	$(0, 0, 0)$	$\{b_1, b_2, b_3\}\{a_1, a_2, a_3\}\{a_4, a_5, a_6\}$
$(1, 1, 2, 2, 3, 2; 0, -1)$	B	$(0, 1, 0)$	$\{b_1, b_3\}\{b_2, a_1, a_2\}\{a_3, a_4, a_6\}\{a_5\}$
$(1, 1, 2, 3, 2, 2; -1, 0)$	D	$(1, 0, 0)$	$\{b_2, b_3\}\{b_1, a_1, a_2\}\{a_3, a_5, a_6\}\{a_4\}$
$(1, 2, 1, 2, 2, 3; 0, -1)$	B	$(0, 1, 0)$	$\{b_1, b_3\}\{b_2, a_1, a_3\}\{a_2, a_4, a_5\}\{a_6\}$
$(1, 2, 1, 2, 3, 2; -1, 0)$	D	$(1, 0, 0)$	$\{b_2, b_3\}\{b_1, a_1, a_3\}\{a_2, a_4, a_6\}\{a_5\}$
$(1, 2, 2, 3, 3, 2; -1, -1)$	E	$(1, 1, 0)$	$\{b_3\}\{b_1, b_2, a_1\}\{a_2, a_3, a_6\}\{a_4, a_5\}$
$(2, 1, 1, 2, 2, 3; -1, 0)$	D	$(1, 0, 0)$	$\{b_2, b_3\}\{b_1, a_2, a_3\}\{a_1, a_4, a_5\}\{a_6\}$
$(2, 1, 2, 3, 2, 3; -1, -1)$	E	$(1, 1, 0)$	$\{b_3\}\{b_1, b_2, a_2\}\{a_1, a_3, a_5\}\{a_4, a_6\}$
$(2, 2, 1, 2, 3, 3; -1, -1)$	E	$(1, 1, 0)$	$\{b_3\}\{b_1, b_2, a_3\}\{a_1, a_2, a_4\}\{a_5, a_6\}$
$(2, 3, 1, 2, 3, 3; -2, -1)$	F	$(2, 1, 0)$	$\{b_3\}\{b_2, a_3\}\{b_1, a_1, a_4\}\{a_2, a_5, a_6\}$
$(3, 1, 2, 2, 3, 3; -2, -1)$	F	$(2, 1, 0)$	$\{b_3\}\{b_2, a_2\}\{b_1, a_3, a_4\}\{a_1, a_5, a_6\}$

References

- [Mtwo] L. Eriksson, *The inverse self-commutator cost at inertia $(m, 2)$: a dimension-free closed formula, block-shift optimizers and a Pieri certificate*, AI Research Record, ARR-2026-1K33A7K90T87AREF (2026). Cited: (2.1)–(2.2), Theorems A–C', Lemma 4.3 (Pieri triples), Corollary 4.6, Section 5.1 (block shifts), Lemma 5.0 (memorylessness), Open question 1.
- [FourLevel] L. Eriksson, *The Exact Four-Level Inverse Commutator Cost: Horn–Littlewood–Richardson Facets, Rank Transitions, and Sharp Loop Synthesis*, AI Research Record, ARR-2026-3M1EEG1T689ADSMW (2026). Cited: Theorem 2.2 (Horn program).
- [FiveLevel] L. Eriksson, *The Exact Five-Level Inverse Commutator Cost: Twelve Horn Chambers, Optimal Rank, and the Sharp $5/2$ Resource Tax*, AI Research Record, ARR-2026-37B8R0QTA894GTFF (2026).
- [OneSpike] L. Eriksson, *One-Spike Inverse Self-Commutators and Exact Three-versus-Four-Kick Curvature Synthesis*, AI Research Record, ARR-2026-7NPRNBW4488HG90K (2026). Cited: Theorem 3.1.
- [RankOnset] L. Eriksson, *Sharp Onset and Unbounded Growth of Norm-Optimal Self-Commutator Rank*, AI Research Record, ARR-2026-5QQF95VHTC9GABH8 (2026). Cited: Lemma 2.1, Proposition 2.2, Remark 3.3 (padding example), Theorem 4.1.
- [RankAdaptive] L. Eriksson, *Sharp Rank-Adaptive Bounds for Inverse Self-Commutators*, AI Research Record, ARR-2026-1D2QV1RP1292JREW (2026). Cited: Theorem 1.1.
- [Ho62] A. Horn, Eigenvalues of sums of Hermitian matrices, *Pacific J. Math.* 12 (1962) 225–241.
- [Kl98] A. A. Klyachko, Stable bundles, representation theory and Hermitian operators, *Selecta*

Math. (N.S.) 4 (1998) 419–445.

- [KT99] A. Knutson, T. Tao, The honeycomb model of $GL_n(\mathbb{C})$ tensor products I: proof of the saturation conjecture, *J. Amer. Math. Soc.* 12 (1999) 1055–1090.
- [Fu00] W. Fulton, Eigenvalues, invariant factors, highest weights, and Schubert calculus, *Bull. Amer. Math. Soc.* 37 (2000) 209–249.
- [Fu97] W. Fulton, *Young Tableaux*, LMS Student Texts 35, Cambridge University Press, 1997 (Pieri rule).
- [Li50] V. B. Lidskii, On the characteristic numbers of the sum and product of symmetric matrices, *Dokl. Akad. Nauk SSSR* 75 (1950) 769–772.
- [Wi55] H. Wielandt, An extremum property of sums of eigenvalues, *Proc. Amer. Math. Soc.* 6 (1955) 106–110.
- [JOS13] W. B. Johnson, N. Ozawa, G. Schechtman, A quantitative version of the commutator theorem for zero trace matrices, *Proc. Natl. Acad. Sci. USA* 110 (2013) 19251–19255.
- [AS15] O. Angel, G. Schechtman, The Hilbert–Schmidt version of the commutator theorem for zero trace matrices, *Bull. London Math. Soc.* 47 (2015) 715–719; arXiv:1503.07980.